In this edge case, it's easy to see that the generic point P_t is given by the vectorial equation:

$$P_L\left(t\right) = E \cdot t + S \cdot \left(1 - t\right), t\epsilon\left[0, 1\right]$$

We won't derive it for a quadratic curve, but here it is how it will look like:

$$P_Q\left(t\right) = E \cdot t^2 + C_1 \cdot 2 \cdot t \cdot \left(1 - t\right) + S \cdot \left(1 - t\right)^2, t\epsilon\left[0, 1\right]$$

In general, for a Bézier curve with n-2 control points, we can express its generic point as:

$$P\left(t\right) = \sum_{i=0}^{n} Q_i \cdot \binom{n}{i} \cdot t^i \cdot \left(1 - t\right)^{n-i}, t\epsilon\left[0, 1\right]$$

using the conventions: $S=Q_0$, $E=Q_n$ and $C_i=Q_{i+1}$ $\forall$ $i=1,..,n-2$.

15.4.4 Intersections Between Quadratic Bézier Curves

In our examples, we are only going to use a subset of the set of Bézier curves: only *symmetric quadratic Bézier curves* where the control point lies exactly at the same distance between the two endpoints. This way, we can simplify the way we reason about the curve, and the way we look for intersections.